

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Nov 2025 P1H Worked Solutions

Nov 2025 | P1H

37 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 4****TOPIC: Statistics Toolkit**

Nov 2025 P1H - Nov 2025 | P1H

- 1** The table gives information about the number of kilometres that Ted cycled on each of the 30 days in April.

Number of kilometres (K)	Frequency
$0 \leq K < 5$	8
$5 \leq K < 10$	7
$10 \leq K < 15$	3
$15 \leq K < 20$	10
$20 \leq K < 25$	2

Calculate an estimate for the mean number of kilometres that Ted cycled on each day in April.

..... kilometres

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION**Q#: 1** **Marks: 4**TOPIC: **Statistics Toolkit**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use midpoints**

Use midpoints 2.5, 7.5, 12.5, 17.5, 22.5.

$$2.5(8) + 7.5(7) + 12.5(3) + 17.5(10) + 22.5(2) = 330$$

$$\text{estimated mean} = \frac{330}{30} = 11$$

FINAL ANSWER

11 km.

EA

PAST PAPER QUESTION**Q#: 2****Marks: 2****TOPIC: Rounding, Estimation & Bounds**

Nov 2025 P1H - Nov 2025 | P1H

- 2 By rounding each number to one significant figure, work out an estimate for the value of

$$\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$$

Show your working clearly.

(Total for Question 2 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 2****Marks: 2****TOPIC: Rounding, Estimation & Bounds**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Round each number to one**

Round each number to one significant figure:

$$2.11 \approx 2, \quad 58.9 \approx 60, \quad 8.859 \approx 9$$

$$\begin{aligned} \frac{2.11^2 \times 58.9}{\sqrt{8.859}} &\approx \frac{2^2 \times 60}{\sqrt{9}} \\ &= \frac{4 \times 60}{3} = 80 \end{aligned}$$

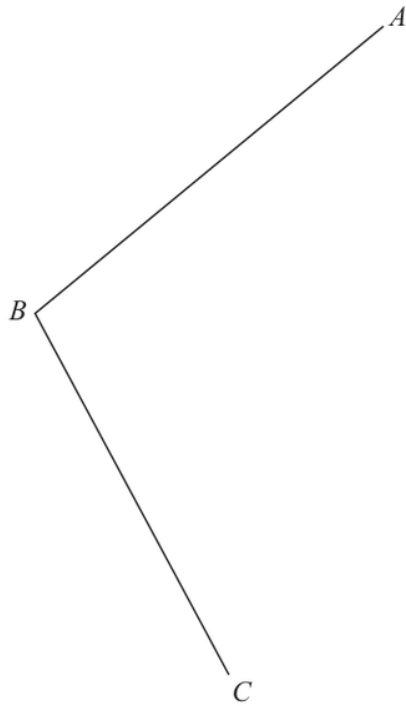
FINAL ANSWER

80

PAST PAPER QUESTION**Q#: 3****Marks: 2****TOPIC: Bearings, Scale Drawing & Constructions**

Nov 2025 P1H - Nov 2025 | P1H

- 3** Using ruler and compasses only, construct the bisector of angle ABC
Show all your construction lines.



(Total for Question 3 is 2 marks)

PAST PAPER SOLUTION

Q#: 3

Marks: 2

TOPIC: Bearings, Scale Drawing & Constructions

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Construction answer

To construct the bisector of angle ABC :

1. Use centre B to draw an arc crossing BA and BC .
2. Use the same compass width from the two crossing points to make two arcs inside the angle.
3. Join B to the point where these two arcs meet.

FINAL ANSWER

That line is the required angle bisector.

EA

PAST PAPER QUESTION**Q#: 4****Marks: 3**TOPIC: **Statistics Toolkit**

Nov 2025 P1H - Nov 2025 | P1H

4 e, f and g represent integers such that

$e \quad f \quad 7 \quad g \quad 12 \quad 17$

is a list of integers written in order of size.

The integers have

- a range of 15
- a median of 8.5
- a mean of 9

Work out the value of e , the value of f and the value of g

$e =$

$f =$

$g =$

(Total for Question 4 is 3 marks)

PAST PAPER SOLUTION**Q#: 4****Marks: 3**TOPIC: **Statistics Toolkit**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The range is**

The range is 15:

$$17 - e = 15$$

$$e = 2$$

2. Calculate statistic

The mean is 9, so the total is

$$6 \times 9 = 54$$

$$2 + f + 7 + g + 12 + 17 = 54$$

$$f + g = 16$$

3. The median is

The median is 8.5:

$$\frac{7 + g}{2} = 8.5$$

$$g = 10$$

$$f = 16 - 10 = 6$$

FINAL ANSWER $e = 2, f = 6, g = 10.$

PAST PAPER QUESTION**Q#: 5****Marks: 3**TOPIC: **Algebraic Proof**

Nov 2025 P1H - Nov 2025 | P1H

5 Show that $3\frac{5}{7} + 1\frac{2}{3} = 5\frac{8}{21}$

(Total for Question 5 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 5****Marks: 3****TOPIC: Algebraic Proof**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Evaluate fraction**

$$\frac{3}{7} + \frac{1}{3}$$

2. Use denominator 21

Use denominator 21:

$$\frac{3}{7} = \frac{9}{21}, \quad \frac{1}{3} = \frac{7}{21}$$

$$\frac{9}{21} + \frac{7}{21} = \frac{16}{21}$$

FINAL ANSWER

$$\frac{16}{21}$$

PAST PAPER QUESTION**Q#: 6****Marks: 4****TOPIC: Ratio Problem Solving**

Nov 2025 P1H - Nov 2025 | P1H

6 Eli and Peta share \$275 in the ratio 2 : 3

Eli gives $\frac{3}{11}$ of his share to charity.

Peta gives 0.32 of her share to charity.

Work out the total amount that Eli and Peta give to charity.

\$.....

(Total for Question 6 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 6

Marks: 4

TOPIC: **Ratio Problem Solving**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The ratio is

The ratio is

$$\text{Eli : Peta} = 2 : 3$$

2. Split the ratio

There are 5 parts, so one part is

$$275 \div 5 = 55$$

3. Eli gets

Eli gets

$$2 \times 55 = 110$$

4. Peta gets

Peta gets

$$3 \times 55 = 165$$

5. Eli gives

Eli gives

$$\frac{3}{11} \times 110 = 30$$

6. Peta gives

Peta gives

$$0.32 \times 165 = 52.8$$

7. Total given to charity

Total given to charity:

$$30 + 52.8 = 82.8$$

FINAL ANSWER

82.8.

PAST PAPER QUESTION**Q#: 7****Marks: 2****TOPIC: Rounding, Estimation & Bounds**

Nov 2025 P1H - Nov 2025 | P1H

7 The weight of a bag of apples is 475 g correct to the nearest g

(a) Write down the lower bound of the weight.

..... g
(1)

The height of a box is 120 cm correct to the nearest 10 cm

(b) Write down the upper bound for the height.

..... cm
(1)

(Total for Question 7 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 7****Marks: 2****TOPIC: Rounding, Estimation & Bounds**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find lower bound**

The weight 475 g correct to the nearest g has lower bound

$$475 - 0.5 = 474.5$$

2. Find upper bound

The height 120 cm correct to the nearest 10 cm has upper bound

$$120 + 5 = 125$$

FINAL ANSWER

(a) 474.5 g, (b) 125 cm

EA

PAST PAPER QUESTION**Q#: 8****Marks: 2**TOPIC: **Algebraic Roots & Indices**

Nov 2025 P1H - Nov 2025 | P1H

8 $\frac{8^{-2} \times 8^9}{8^{10}} = 8^n$

Work out the value of n $n = \dots\dots\dots$

(Total for Question 8 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 8****Marks: 2**TOPIC: **Algebraic Roots & Indices**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use index laws**

$$\frac{8^{-2} \times 8^9}{8^{10}} = 8^{-2+9-10} = 8^{-3}$$

FINAL ANSWER

$$n = -3$$

EA

PAST PAPER QUESTION**Q#: 9****Marks: 3****TOPIC: Percentages**

Nov 2025 P1H - Nov 2025 | P1H

- 9** Hari scored 140 points playing a game on Tuesday.
This was 12% more points than he scored playing the game on Monday.
Work out the number of points that Hari scored playing the game on Monday.

(Total for Question 9 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 9****Marks: 3**TOPIC: **Percentages**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Tuesday s score was more**

Tuesday's score was 12% more than Monday's score, so Tuesday was 112% of Monday.

$$1.12 \times \text{Monday} = 140$$

$$\text{Monday} = \frac{140}{1.12} = 125$$

FINAL ANSWER

125 points

EA

PAST PAPER QUESTION**Q#: 10****Marks: 7****TOPIC: Factorising**

Nov 2025 P1H - Nov 2025 | P1H

10 (a) Simplify $(6m)^0$
(1)(b) Solve $5y + 20 < 7y + 1$
(2)(c) Factorise fully $15w^2x^5 + 25w^3x^2$
(2)(d) Simplify fully $\frac{36a^5c^7}{12a^2c^3}$
(2)**(Total for Question 10 is 7 marks)**

PAST PAPER SOLUTION**Q#: 10****Marks: 7****TOPIC: Factorising**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a) $(6m)^0 = 1.$

2. (b)

(b)

$$5y + 20 < 7y + 1$$

$$19 < 2y$$

$$y > \frac{19}{2}$$

3. (c)

(c)

$$15w^2x^5 + 25w^3x^2 = 5w^2x^2(3x^3 + 5w)$$

4. (d)

(d)

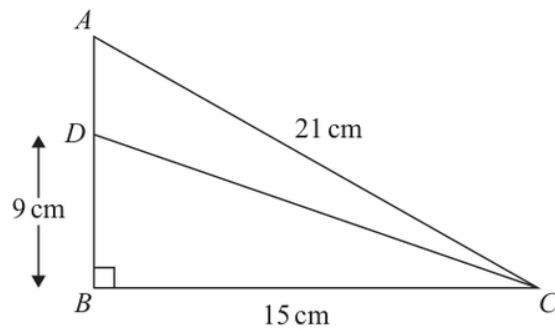
$$\frac{36a^5c^7}{12a^2c^3} = 3a^3c^4$$

FINAL ANSWER

$$1, y > \frac{19}{2}, 5w^2x^2(3x^3 + 5w), 3a^3c^4$$

PAST PAPER QUESTION**Q#: 11****Marks: 4****TOPIC:** Right-Angled Triangles - Pythagoras & Trigonometry

Nov 2025 P1H - Nov 2025 | P1H

11 ABC is a right-angled triangle.Diagram **NOT**
accurately drawn

$$AC = 21 \text{ cm} \quad BC = 15 \text{ cm} \quad \text{angle } ABC = 90^\circ$$

The point D lies on AB such that $DB = 9 \text{ cm}$ Work out the size of angle ACD

Give your answer correct to one decimal place.

o

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION**Q#: 11****Marks: 4****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**First find AB :

$$AB^2 = 21^2 - 15^2 = 216$$

$$AB = 14.696 \dots$$

$$AD = AB - DB = 14.696 \dots - 9 = 5.696 \dots$$

2. AngleAngle ACB :

$$\cos ACB = \frac{15}{21}$$

$$ACB = 44.415 \dots^\circ$$

3. AngleAngle DCB :

$$\tan DCB = \frac{9}{15}$$

$$DCB = 30.963 \dots^\circ$$

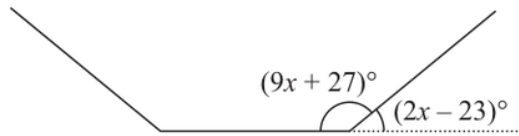
$$ACD = 44.415 \dots - 30.963 \dots = 13.451 \dots^\circ$$

FINAL ANSWER

13.5°.

PAST PAPER QUESTION**Q#: 12****Marks: 4****TOPIC: Angles in Polygons & Parallel Lines**

Nov 2025 P1H - Nov 2025 | P1H

12Diagram **NOT**
accurately drawnThe diagram shows part of a regular n -sided polygon withan interior angle of $(9x + 27)^\circ$ an exterior angle of $(2x - 23)^\circ$ Work out the value of n $n = \dots\dots\dots$ **(Total for Question 12 is 4 marks)**

EA

PAST PAPER SOLUTION**Q#: 12****Marks: 4****TOPIC: Angles in Polygons & Parallel Lines**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The interior angle and exterior**

The interior angle and exterior angle of a regular polygon add to 180° :

$$(9x + 27) + (2x - 23) = 180$$

$$11x + 4 = 180$$

$$x = 16$$

2. The exterior angle is

The exterior angle is

$$2x - 23 = 2(16) - 23 = 9^\circ$$

3. Find the gradient

For a regular polygon,

$$n = \frac{360}{9} = 40$$

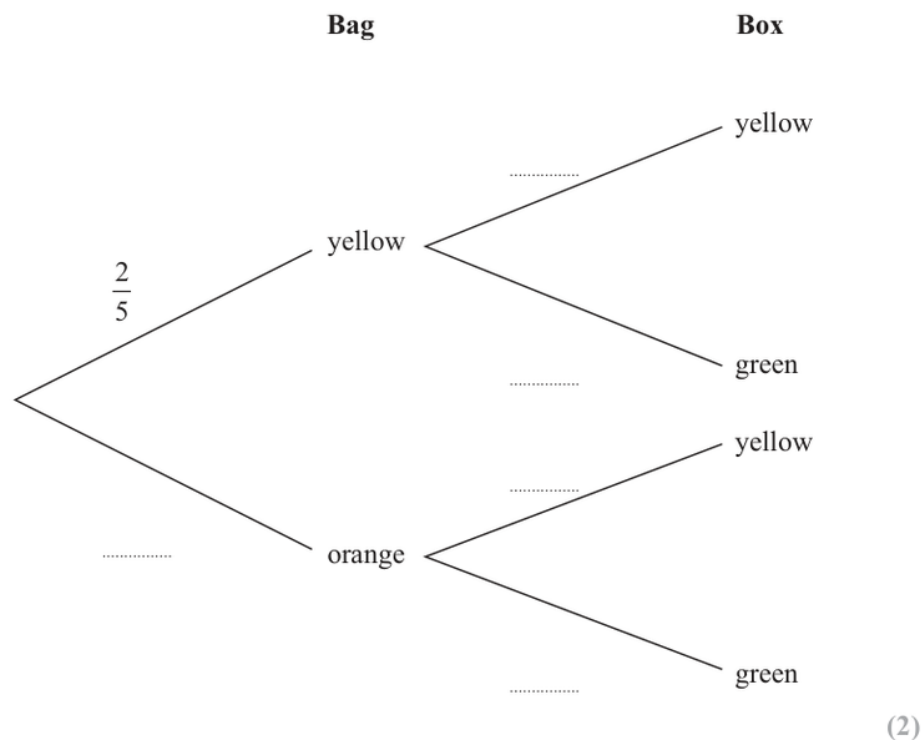
FINAL ANSWER

$$n = 40.$$

PAST PAPER QUESTION**Q#: 13****Marks: 5****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Nov 2025 P1H - Nov 2025 | P1H

- 13** In a bag, there are only 2 yellow beads and 3 orange beads.
 In a box, there are only 3 yellow beads and 5 green beads.
- Chi takes at random a bead from the bag and a bead from the box.
- (a) Complete the probability tree diagram.



- (b) Work out the probability that Chi takes two beads of different colours.

.....

(3)

(Total for Question 13 is 5 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 5****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Bag 1**

Bag 1:

$$P(Y) = \frac{2}{5}, \quad P(O) = \frac{3}{5}$$

2. Bag 2

Bag 2:

$$P(Y) = \frac{3}{8}, \quad P(G) = \frac{5}{8}$$

3. The only way to get

The only way to get the same colour is yellow then yellow.

$$\begin{aligned} P(\text{different}) &= 1 - \frac{2}{5} \times \frac{3}{8} \\ &= 1 - \frac{3}{20} = \frac{17}{20} \end{aligned}$$

FINAL ANSWER

$$\frac{17}{20}$$

PAST PAPER QUESTION**Q#: 14****Marks: 6****TOPIC: Expanding Brackets**

Nov 2025 P1H - Nov 2025 | P1H

14 (a) Expand and simplify $7x(3x + 2)(2x - 5)$
(3)(b) Solve $\frac{9}{2y} + \frac{5}{7} = 5$

Show clear algebraic working.

 $y =$
(3)

(Total for Question 14 is 6 marks)

EA

PAST PAPER SOLUTION**Q#: 14****Marks: 6****TOPIC: Expanding Brackets**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$\begin{aligned}7x(3x + 2)(2x - 5) &= 7x(6x^2 - 11x - 10) \\ &= 42x^3 - 77x^2 - 70x\end{aligned}$$

2. (b)

(b)

$$\frac{9}{2y} + \frac{5}{7} = 5$$

$$\frac{9}{2y} = \frac{30}{7}$$

$$63 = 60y$$

FINAL ANSWER(a) $42x^3 - 77x^2 - 70x$, (b) $y = \frac{21}{20}$

PAST PAPER QUESTION**Q#: 15****Marks: 2****TOPIC: Circles, Arcs & Sectors**

Nov 2025 P1H - Nov 2025 | P1H

15 AOB is a sector of a circle with centre O and radius 12 cm

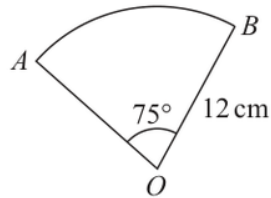


Diagram **NOT**
accurately drawn

Work out the area of the sector.

Give your answer correct to 3 significant figures.

..... cm²

(Total for Question 15 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 15****Marks: 2****TOPIC: Circles, Arcs & Sectors**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate area**

$$\begin{aligned}\text{area} &= \frac{75}{360} \times \pi(12)^2 \\ &= 94.247 \dots\end{aligned}$$

FINAL ANSWER94.2 cm².

EA

PAST PAPER QUESTION**Q#: 16****Marks: 2****TOPIC: Statistics Toolkit**

Nov 2025 P1H - Nov 2025 | P1H

16 Ivan asked 15 people how many books they read last year.

Here are his results.

1 1 3 4 6 7 8 9 10 12 15 25 30 37 50

Work out the interquartile range of the number of books read.

(Total for Question 16 is 2 marks)

EA

PAST PAPER SOLUTION**Q#: 16****Marks: 2****TOPIC: Statistics Toolkit**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use Statistics Toolkit correct**

The data are already in order:

 $1, 1, 3, 4, 6, 7, 8, 9, 10, 12, 15, 25, 30, 37, 50$ **2. There are 15 values**

There are 15 values.

$$Q_1 = 4, \quad Q_3 = 25$$

$$\text{IQR} = 25 - 4 = 21$$

FINAL ANSWER

21.

PAST PAPER QUESTION**Q#: 17****Marks: 3****TOPIC: Algebraic Proof**

Nov 2025 P1H - Nov 2025 | P1H

17 Show that $\frac{2\sqrt{7} + 2}{\sqrt{7} - 3}$ can be written in the form $a - \sqrt{b}$ where a and b are integers.

Show your working clearly.

(Total for Question 17 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 17****Marks: 3****TOPIC: Algebraic Proof**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify surd**

$$\begin{aligned}
 \frac{2\sqrt{7}+2}{\sqrt{7}-3} \times \frac{\sqrt{7}+3}{\sqrt{7}+3} &= \frac{(2\sqrt{7}+2)(\sqrt{7}+3)}{7-9} \\
 &= \frac{14+6\sqrt{7}+2\sqrt{7}+6}{-2} \\
 &= \frac{20+8\sqrt{7}}{-2} = -10-4\sqrt{7}
 \end{aligned}$$

2. Simplify surdSince $4\sqrt{7} = \sqrt{112}$,

$$-10-4\sqrt{7} = -10-\sqrt{112}$$

FINAL ANSWER $-10-\sqrt{112}$, so $a = -10$ and $b = 112$.

PAST PAPER QUESTION**Q#: 18****Marks: 4****TOPIC: Rearranging Formulas**

Nov 2025 P1H - Nov 2025 | P1H

18 Make y the subject of $p = \sqrt{\frac{y+w}{3y-t}}$

(Total for Question 18 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 18****Marks: 4****TOPIC: Rearranging Formulas**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Rearrange formula**

$$p = \sqrt{\frac{y+w}{3y-t}}$$

$$p^2 = \frac{y+w}{3y-t}$$

$$p^2(3y-t) = y+w$$

$$3p^2y - p^2t = y+w$$

$$y(3p^2 - 1) = w + p^2t$$

FINAL ANSWER

$$y = \frac{w+p^2t}{3p^2-1}$$

PAST PAPER QUESTION**Q#: 19****Marks: 3****TOPIC: Sine, Cosine Rule & Area of Triangles**

Nov 2025 P1H - Nov 2025 | P1H

19 The diagram shows triangle ABC

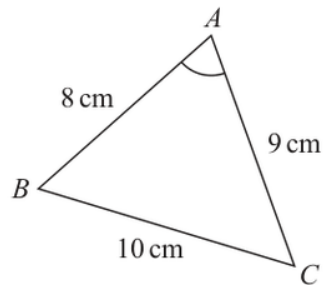


Diagram **NOT**
accurately drawn

$$AB = 8 \text{ cm} \quad BC = 10 \text{ cm} \quad CA = 9 \text{ cm}$$

Work out the size of angle BAC

Give your answer correct to one decimal place.

o

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION**Q#: 19****Marks: 3****TOPIC: Sine, Cosine Rule & Area of Triangles**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use cosine rule**Use the cosine rule for angle BAC :

$$\cos A = \frac{8^2 + 9^2 - 10^2}{2(8)(9)}$$

$$\cos A = \frac{45}{144}$$

$$A = 71.790 \dots^\circ$$

FINAL ANSWER

71.8°.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Coordinate Geometry**

Nov 2025 P1H - Nov 2025 | P1H

20 The straight line **L** has equation $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**

Give your answer in the form $y = mx + c$

(Total for Question 20 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 3****TOPIC: Coordinate Geometry**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Line has equation**Line L has equation

$$y = 4x + 7$$

2. Find the gradient

So its gradient is

$$4$$

3. Find the gradientLine M is perpendicular to L , so its gradient is

$$-\frac{1}{4}$$

4. Evaluate fractionLine M passes through $(8, 1)$.

$$y - 1 = -\frac{1}{4}(x - 8)$$

$$y - 1 = -\frac{1}{4}x + 2$$

$$y = -\frac{1}{4}x + 3$$

FINAL ANSWER

$$y = -\frac{1}{4}x + 3$$

PAST PAPER QUESTION**Q#: 20****Marks: 3****TOPIC: Coordinate Geometry**

Nov 2025 P1H - Nov 2025 | P1H

20 The straight line **L** has equation $y = 4x + 7$

The straight line **M** is perpendicular to **L** and passes through the point with coordinates (8, 1)

Find an equation for **M**

Give your answer in the form $y = mx + c$

(Total for Question 20 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 3****TOPIC: Coordinate Geometry**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Line has equation**Line L has equation

$$y = 4x + 7$$

2. Find the gradient

So its gradient is

$$4$$

3. Find the gradientLine M is perpendicular to L , so its gradient is

$$-\frac{1}{4}$$

4. Evaluate fractionLine M passes through $(8, 1)$.

$$y - 1 = -\frac{1}{4}(x - 8)$$

$$y - 1 = -\frac{1}{4}x + 2$$

$$y = -\frac{1}{4}x + 3$$

FINAL ANSWER

$$y = -\frac{1}{4}x + 3$$

PAST PAPER QUESTION**Q#: 21****Marks: 3****TOPIC: Completing the Square**

Nov 2025 P1H - Nov 2025 | P1H

21 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a , b and c are integers.

(Total for Question 21 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 21** **Marks: 3****TOPIC: Completing the Square**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve inequality**

$$5x^2 - 20x + 23 = 5(x^2 - 4x) + 23$$

$$= 5((x - 2)^2 - 4) + 23$$

$$= 5(x - 2)^2 - 20 + 23$$

$$= 5(x - 2)^2 + 3$$

FINAL ANSWER

$$5(x - 2)^2 + 3$$

PAST PAPER QUESTION**Q#: 21****Marks: 3****TOPIC: Completing the Square**

Nov 2025 P1H - Nov 2025 | P1H

21 Express $5x^2 - 20x + 23$ in the form $a(x - b)^2 + c$ where a , b and c are integers.

(Total for Question 21 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 21****Marks: 3****TOPIC: Completing the Square**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve inequality**

$$5x^2 - 20x + 23 = 5(x^2 - 4x) + 23$$

$$= 5((x - 2)^2 - 4) + 23$$

$$= 5(x - 2)^2 - 20 + 23$$

$$= 5(x - 2)^2 + 3$$

FINAL ANSWER

$$5(x - 2)^2 + 3$$

PAST PAPER QUESTION**Q#: 22****Marks: 3****TOPIC: Combined & Conditional Probability**

Nov 2025 P1H - Nov 2025 | P1H

22 There are 15 buttons in a box.

3 of the buttons are red

2 of the buttons are pink

10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

(Total for Question 22 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 22****Marks: 3****TOPIC: Combined & Conditional Probability**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. There are 2 pink buttons**

There are 2 pink buttons.

2. At least one pink button

At least one pink button is still in the box unless both pink buttons are taken.

3. Use the complement

So use the complement.

4. The probability that both pink

The probability that both pink buttons are taken is

$$\begin{aligned} & \frac{\binom{2}{2} \binom{13}{1}}{\binom{15}{3}} \\ &= \frac{13}{455} \\ &= \frac{1}{35} \end{aligned}$$

5. Evaluate fraction

Therefore,

$$\begin{aligned} P(\text{at least one pink is still in the box}) &= 1 - \frac{1}{35} \\ &= \frac{34}{35} \end{aligned}$$

FINAL ANSWER

$$\frac{34}{35}$$

PAST PAPER QUESTION**Q#: 22****Marks: 3****TOPIC: Combined & Conditional Probability**

Nov 2025 P1H - Nov 2025 | P1H

22 There are 15 buttons in a box.

3 of the buttons are red

2 of the buttons are pink

10 of the buttons are blue

Pete takes at random three buttons from the box.

Work out the probability that there is still at least one pink button in the box.

(Total for Question 22 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 22****Marks: 3****TOPIC: Combined & Conditional Probability**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. There are 2 pink buttons**

There are 2 pink buttons.

2. At least one pink button

At least one pink button is still in the box unless both pink buttons are taken.

3. Use the complement

So use the complement.

4. The probability that both pink

The probability that both pink buttons are taken is

$$\begin{aligned} & \frac{\binom{2}{2} \binom{13}{1}}{\binom{15}{3}} \\ &= \frac{13}{455} \\ &= \frac{1}{35} \end{aligned}$$

5. Evaluate fraction

Therefore,

$$\begin{aligned} P(\text{at least one pink is still in the box}) &= 1 - \frac{1}{35} \\ &= \frac{34}{35} \end{aligned}$$

FINAL ANSWER

$$\frac{34}{35}$$

PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Algebraic Fractions**

Nov 2025 P1H - Nov 2025 | P1H

23 $a = \frac{2x+5}{1-x}$ $x = \frac{5-2y}{3y}$

Write a in the form $\frac{m+ny}{p(y-1)}$ where m , n and p are integers.

Show your working clearly.

$a =$

(Total for Question 23 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 23****Marks: 3****TOPIC: Algebraic Fractions**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Split the ratio**

$$a = \frac{2x + 5}{1 - x}, \quad x = \frac{5 - 2y}{3y}$$

2. Split the ratioSubstitute for x .**3. Numerator**

Numerator:

$$\begin{aligned} 2x + 5 &= 2 \left(\frac{5 - 2y}{3y} \right) + 5 \\ &= \frac{10 - 4y + 15y}{3y} = \frac{10 + 11y}{3y} \end{aligned}$$

4. Denominator

Denominator:

$$\begin{aligned} 1 - x &= 1 - \frac{5 - 2y}{3y} \\ &= \frac{3y - 5 + 2y}{3y} = \frac{5y - 5}{3y} = \frac{5(y - 1)}{3y} \end{aligned}$$

5. Split the ratio

Therefore

$$a = \frac{\frac{10 + 11y}{3y}}{\frac{5(y - 1)}{3y}} = \frac{10 + 11y}{5(y - 1)}$$

FINAL ANSWER

$$\frac{10 + 11y}{5(y - 1)}$$

PAST PAPER QUESTION**Q#: 23****Marks: 3****TOPIC: Algebraic Fractions**

Nov 2025 P1H - Nov 2025 | P1H

23 $a = \frac{2x+5}{1-x}$ $x = \frac{5-2y}{3y}$

Write a in the form $\frac{m+ny}{p(y-1)}$ where m , n and p are integers.

Show your working clearly.

$a =$

(Total for Question 23 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 23****Marks: 3****TOPIC: Algebraic Fractions**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Split the ratio**

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Numerator:

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Therefore

$$a = \frac{\frac{10 + 11y}{3y}}{\frac{5(y - 1)}{3y}} = \frac{10 + 11y}{5(y - 1)}$$

FINAL ANSWER

$$\frac{10 + 11y}{5(y - 1)}$$

PAST PAPER QUESTION**Q#: 24****Marks: 3****TOPIC: Transformations of Graphs**

Nov 2025 P1H - Nov 2025 | P1H

24 A curve has equation $y = f(x)$

There is only one turning point on the curve.

The coordinates of this turning point are (4, 3)

Write down the coordinates of the turning point on the curve with equation

(i) $y = f(x + 5)$

(..... ,)
(1)

(ii) $y = f(x) + 7$

(..... ,)
(1)

(iii) $y = f(2x)$

(..... ,)
(1)**(Total for Question 24 is 3 marks)**

PAST PAPER SOLUTION**Q#: 24****Marks: 3****TOPIC: Transformations of Graphs**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The turning point of is**The turning point of $y = f(x)$ is $(4, 3)$.**2. Calculate value**

For

$$y = f(x + 5)$$

3. the graph moves 5 units

the graph moves 5 units left:

$$(4, 3) \rightarrow (-1, 3)$$

4. Calculate value

For

$$y = f(x) + 7$$

5. the graph moves 7 units

the graph moves 7 units up:

$$(4, 3) \rightarrow (4, 10)$$

6. Calculate value

For

$$y = f(2x)$$

7. the coordinate is divided bythe x -coordinate is divided by 2:

$$(4, 3) \rightarrow (2, 3)$$

FINAL ANSWER $(-1, 3)$, $(4, 10)$, and $(2, 3)$

PAST PAPER QUESTION**Q#: 24****Marks: 3****TOPIC: Transformations of Graphs**

Nov 2025 P1H - Nov 2025 | P1H

24 A curve has equation $y = f(x)$

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(1)**(Total for Question 24 is 3 marks)**

PAST PAPER SOLUTION**Q#: 24****Marks: 3****TOPIC: Transformations of Graphs**

Nov 2025 P1H - Nov 2025 | P1H

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the graph moves 7 units up:

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For

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7. the coordinate is divided bythe x -coordinate is divided by 2:

$$(4, 3) \rightarrow (2, 3)$$

FINAL ANSWER $(-1, 3)$, $(4, 10)$, and $(2, 3)$

PAST PAPER QUESTION**Q#: 25****Marks: 3****TOPIC: Solving Inequalities**

Nov 2025 P1H - Nov 2025 | P1H

- 25** Solve the inequality $2x^2 + x - 28 > 0$
Show clear algebraic working.

(Total for Question 25 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 25****Marks: 3****TOPIC: Solving Inequalities**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve inequality**

$$2x^2 + x - 28 > 0$$

2. Factorise

Factorise:

$$2x^2 + x - 28 = (2x - 7)(x + 4)$$

3. The critical values are

The critical values are

$$x = \frac{7}{2} \quad \text{and} \quad x = -4$$

4. The quadratic is greater than

The quadratic is greater than 0 outside the roots.

FINAL ANSWER

$$x < -4 \text{ or } x > \frac{7}{2}$$

PAST PAPER QUESTION**Q#: 25****Marks: 3**TOPIC: **Solving Inequalities**

Nov 2025 P1H - Nov 2025 | P1H

- 25** Solve the inequality $2x^2 + x - 28 > 0$
Show clear algebraic working.

(Total for Question 25 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 25****Marks: 3****TOPIC: Solving Inequalities**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve inequality**

$$2x^2 + x - 28 > 0$$

2. Factorise

Factorise:

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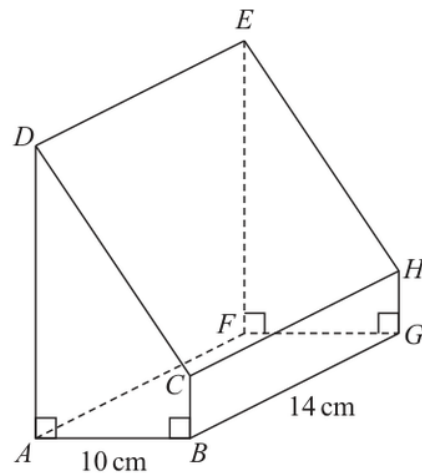
The quadratic is greater than 0 outside the roots.

FINAL ANSWER

$$x < -4 \text{ or } x > \frac{7}{2}$$

PAST PAPER QUESTION**Q#: 26****Marks: 5****TOPIC: 3D Pythagoras & Trigonometry**

Nov 2025 P1H - Nov 2025 | P1H

26 Here is a prism $ABCDEFGH$ with a horizontal, rectangular base $ABGF$ Diagram **NOT**
accurately drawn

$$AB = 10 \text{ cm} \quad BG = 14 \text{ cm}$$

$$\text{angle } DAB = \text{angle } ABC = \text{angle } EFG = \text{angle } FGH = 90^\circ$$

$$BC = \frac{1}{5} AD$$

The angle of elevation of E from B is 50°

Calculate the volume of the prism.

Give your answer correct to 3 significant figures.

Show your working clearly.

..... cm^3

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION

Q#: 26

Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The horizontal base is a

The horizontal base $ABGF$ is a rectangle.

$$AB = 10, \quad BG = 14$$

2. Use trigonometry

So the horizontal distance from B to F is

$$BF = \sqrt{10^2 + 14^2} = \sqrt{296}$$

3. Use trigonometry

The angle of elevation of E from B is 50° .

4. Use trigonometry

Since E is vertically above F ,

$$\tan 50^\circ = \frac{FE}{BF}$$

$$FE = \sqrt{296} \tan 50^\circ$$

5. The prism has the same

The prism has the same cross section throughout, so

$$AD = FE = \sqrt{296} \tan 50^\circ$$

6. Use trigonometry

Also

$$BC = \frac{1}{5}AD$$

7. Area of the trapezium cross

Area of the trapezium cross section $ABCD$:

$$\begin{aligned} & \frac{1}{2}(AD + BC) \times AB \\ &= \frac{1}{2} \left(AD + \frac{1}{5}AD \right) \times 10 \\ &= 6AD \end{aligned}$$

8. Volume of the prism

Volume of the prism:

$$6AD \times 14 = 84AD$$

$$= 84\sqrt{296} \tan 50^\circ$$

$$= 1722.311 \dots$$

9. Correct to 3 significant figures

Correct to 3 significant figures:

$$1720$$

FINAL ANSWER

$$1720 \text{ cm}^3$$

EA

PAST PAPER QUESTION

Q#: 26

Marks: 5

TOPIC: **3D Pythagoras & Trigonometry**

Nov 2025 P1H - Nov 2025 | P1H

26 Here is a prism $ABCDEFGH$ with a horizontal, rectangular base $ABGF$

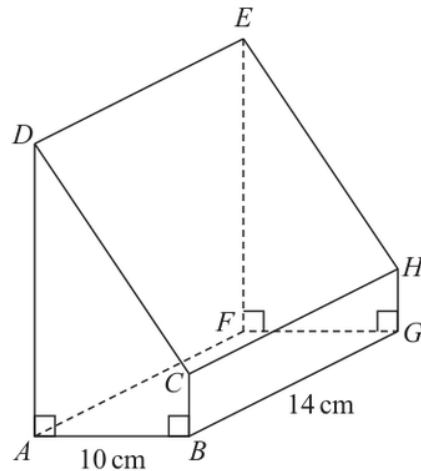


Diagram **NOT**
accurately drawn

$$AB = 10 \text{ cm} \quad BG = 14 \text{ cm}$$

$$\text{angle } DAB = \text{angle } ABC = \text{angle } EFG = \text{angle } FGH = 90^\circ$$

$$BC = \frac{1}{5} AD$$

The angle of elevation of E from B is 50°

Calculate the volume of the prism.
Give your answer correct to 3 significant figures.
Show your working clearly.

..... cm^3

(Total for Question 26 is 5 marks)

PAST PAPER SOLUTION**Q#: 26****Marks: 5****TOPIC: 3D Pythagoras & Trigonometry**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The horizontal base is a**

The horizontal base $ABGF$ is a rectangle.

$$AB = 10, \quad BG = 14$$

2. Use trigonometry

So the horizontal distance from B to F is

$$BF = \sqrt{10^2 + 14^2} = \sqrt{296}$$

3. Use trigonometry

The angle of elevation of E from B is 50° .

4. Use trigonometry

Since E is vertically above F ,

$$\tan 50^\circ = \frac{FE}{BF}$$

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The prism has the same cross section throughout, so

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Area of the trapezium cross section $ABCD$:

$$\begin{aligned} & \frac{1}{2}(AD + BC) \times AB \\ &= \frac{1}{2} \left(AD + \frac{1}{5}AD \right) \times 10 \\ &= 6AD \end{aligned}$$

8. Volume of the prism

Volume of the prism:

$$6AD \times 14 = 84AD$$

$$= 84\sqrt{296} \tan 50^\circ$$

$$= 1722.311 \dots$$

9. Correct to 3 significant figures

Correct to 3 significant figures:

$$1720$$

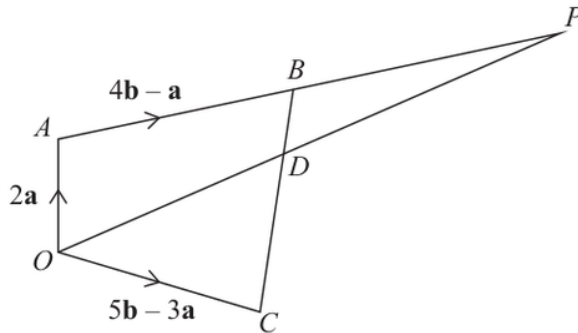
FINAL ANSWER

$$1720 \text{ cm}^3$$

EA

PAST PAPER QUESTION**Q#: 27****Marks: 6****TOPIC: Vectors**

Nov 2025 P1H - Nov 2025 | P1H

27Diagram **NOT**
accurately drawn

$OABC$ is a quadrilateral.
 ABP and ODP are straight lines.

$$\vec{OA} = 2\mathbf{a} \quad \vec{AB} = 4\mathbf{b} - \mathbf{a} \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

- (a) Find an expression in terms of $\mathbf{a}$ and $\mathbf{b}$ for the vector $\vec{BC}$
 Simplify your answer.

.....
(2)

The point D lies on BC such that $BD:DC = 1:3$

Given that $\vec{OP} = n\vec{OD}$

- (b) use a vector method to find the value of n

$n =$
(4)

(Total for Question 27 is 6 marks)

PAST PAPER SOLUTION

Q#: 27

Marks: 6

TOPIC: **Vectors**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Read the graph

$$\vec{OA} = 2\mathbf{a}, \quad \vec{AB} = 4\mathbf{b} - \mathbf{a}, \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

2. Read the graphFirst find $\vec{OB}$:

$$\vec{OB} = \vec{OA} + \vec{AB}$$

$$\vec{OB} = 2\mathbf{a} + 4\mathbf{b} - \mathbf{a}$$

$$\vec{OB} = \mathbf{a} + 4\mathbf{b}$$

3. Read the graph

For part (a),

$$\vec{BC} = \vec{OC} - \vec{OB}$$

$$\vec{BC} = (5\mathbf{b} - 3\mathbf{a}) - (\mathbf{a} + 4\mathbf{b})$$

$$\vec{BC} = \mathbf{b} - 4\mathbf{a}$$

4. Read the graphFor part (b), $BD : DC = 1 : 3$, so

$$\vec{OD} = \vec{OB} + \frac{1}{4}\vec{BC}$$

$$\vec{OD} = \mathbf{a} + 4\mathbf{b} + \frac{1}{4}(\mathbf{b} - 4\mathbf{a})$$

$$\vec{OD} = \frac{17}{4}\mathbf{b}$$

5. Read the graphSince A, B, P are collinear, let

$$\vec{OP} = \vec{OA} + s\vec{AB}$$

$$\vec{OP} = 2\mathbf{a} + s(4\mathbf{b} - \mathbf{a})$$

$$\vec{OP} = (2 - s)\mathbf{a} + 4s\mathbf{b}$$

6. Read the graph

But O, D, P are collinear, and $\vec{OD}$ has only a $\mathbf{b}$ component.

7. Read the graph

So

$$2 - s = 0$$

$$s = 2$$

8. Read the graph

Then

$$\vec{OP} = 8\mathbf{b}$$

9. Given

Given

$$\vec{OP} = n\vec{OD}$$

$$8\mathbf{b} = n\left(\frac{17}{4}\mathbf{b}\right)$$

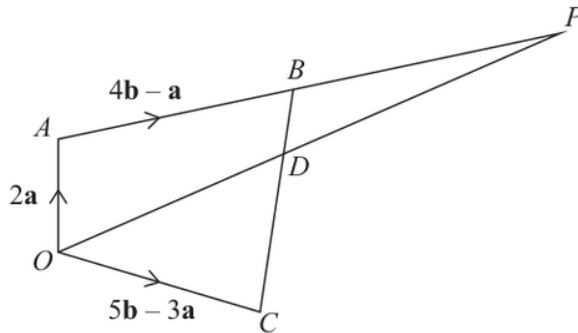
$$n = \frac{32}{17}$$

FINAL ANSWER

$$\vec{BC} = \mathbf{b} - 4\mathbf{a}, \quad n = \frac{32}{17}$$

PAST PAPER QUESTION**Q#: 27****Marks: 6****TOPIC: Vectors**

Nov 2025 P1H - Nov 2025 | P1H

27Diagram **NOT**
accurately drawn

$OABC$ is a quadrilateral.
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$$\vec{OA} = 2\mathbf{a} \quad \vec{AB} = 4\mathbf{b} - \mathbf{a} \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

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 Simplify your answer.

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The point D lies on BC such that $BD:DC = 1:3$

Given that $\vec{OP} = n\vec{OD}$

- (b) use a vector method to find the value of n

$n =$
(4)

(Total for Question 27 is 6 marks)

PAST PAPER SOLUTION**Q#: 27****Marks: 6**TOPIC: **Vectors**

Nov 2025 P1H - Nov 2025 | P1H

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$$\vec{OA} = 2\mathbf{a}, \quad \vec{AB} = 4\mathbf{b} - \mathbf{a}, \quad \vec{OC} = 5\mathbf{b} - 3\mathbf{a}$$

2. Read the graphFirst find $\vec{OB}$:

$$\vec{OB} = \vec{OA} + \vec{AB}$$

$$\vec{OB} = 2\mathbf{a} + 4\mathbf{b} - \mathbf{a}$$

$$\vec{OB} = \mathbf{a} + 4\mathbf{b}$$

3. Read the graph

For part (a),

$$\vec{BC} = \vec{OC} - \vec{OB}$$

$$\vec{BC} = (5\mathbf{b} - 3\mathbf{a}) - (\mathbf{a} + 4\mathbf{b})$$

$$\vec{BC} = \mathbf{b} - 4\mathbf{a}$$

4. Read the graphFor part (b), $BD : DC = 1 : 3$, so

$$\vec{OD} = \vec{OB} + \frac{1}{4}\vec{BC}$$

$$\vec{OD} = \mathbf{a} + 4\mathbf{b} + \frac{1}{4}(\mathbf{b} - 4\mathbf{a})$$

$$\vec{OD} = \frac{17}{4}\mathbf{b}$$

5. Read the graphSince A, B, P are collinear, let

$$\vec{OP} = \vec{OA} + s\vec{AB}$$

$$\vec{OP} = 2\mathbf{a} + s(4\mathbf{b} - \mathbf{a})$$

$$\vec{OP} = (2 - s)\mathbf{a} + 4s\mathbf{b}$$

6. Read the graph

But O, D, P are collinear, and $\vec{OD}$ has only a $\mathbf{b}$ component.

7. Read the graph

So

$$2 - s = 0$$

$$s = 2$$

8. Read the graph

Then

$$\vec{OP} = 8\mathbf{b}$$

9. Given

Given

$$\vec{OP} = n\vec{OD}$$

$$8\mathbf{b} = n\left(\frac{17}{4}\mathbf{b}\right)$$

$$n = \frac{32}{17}$$

FINAL ANSWER

$$\vec{BC} = \mathbf{b} - 4\mathbf{a}, \quad n = \frac{32}{17}$$

PAST PAPER QUESTION**Q#: 28****Marks: 6****TOPIC: Coordinate Geometry**

Nov 2025 P1H - Nov 2025 | P1H

- 28** The curve **C** has equation $x^2 + y^2 = d - 11x$ where d is an integer.
The line **L** has equation $y = 3x + e$ where e is an integer.

C and **L** intersect at the point A and at the point B

The coordinates of A are $(0.2, 2.6)$

Work out the coordinates of B

Show your working clearly.

(..... ,)

(Total for Question 28 is 6 marks)

EA

PAST PAPER SOLUTION**Q#: 28****Marks: 6****TOPIC: Coordinate Geometry**

Nov 2025 P1H - Nov 2025 | P1H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The line is**

The line is

$$y = 3x + e$$

2. Point lies on the linePoint $A = (0.2, 2.6)$ lies on the line.

$$2.6 = 3(0.2) + e$$

$$2.6 = 0.6 + e$$

$$e = 2$$

3. Calculate value

So

$$y = 3x + 2$$

4. The curve is

The curve is

$$x^2 + y^2 = d - 11x$$

5. Use pointUse point $A = (0.2, 2.6)$:

$$0.2^2 + 2.6^2 = d - 11(0.2)$$

$$0.04 + 6.76 = d - 2.2$$

$$d = 9$$

6. Substitute into the curveNow substitute $y = 3x + 2$ into the curve:

$$x^2 + (3x + 2)^2 = 9 - 11x$$

$$x^2 + 9x^2 + 12x + 4 = 9 - 11x$$

$$10x^2 + 23x - 5 = 0$$

7. One root is

One root is $0.2 = \frac{1}{5}$.

8. The product of the roots

The product of the roots is

$$\frac{-5}{10} = -\frac{1}{2}$$

9. Evaluate fraction

So the other root is

$$\frac{-\frac{1}{2}}{\frac{1}{5}} = -\frac{5}{2}$$

10. Evaluate fraction

Then

$$y = 3\left(-\frac{5}{2}\right) + 2$$

$$y = -\frac{11}{2}$$

FINAL ANSWER

$$B = \left(-\frac{5}{2}, -\frac{11}{2}\right)$$

PAST PAPER QUESTION**Q#: 28****Marks: 6****TOPIC: Coordinate Geometry**

Nov 2025 P1H - Nov 2025 | P1H

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C and **L** intersect at the point A and at the point B

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Work out the coordinates of B
Show your working clearly.

(..... ,)

(Total for Question 28 is 6 marks)

PAST PAPER SOLUTION**Q#: 28****Marks: 6**TOPIC: **Coordinate Geometry**

Nov 2025 P1H - Nov 2025 | P1H

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So the other root is

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10. Evaluate fraction

Then

$$y = 3 \left(-\frac{5}{2} \right) + 2$$

$$y = -\frac{11}{2}$$

FINAL ANSWER

$$B = \left(-\frac{5}{2}, -\frac{11}{2} \right)$$